10) Lasso Regression

```
import numpy as np
from sklearn.datasets import
load_diabetes
from sklearn.linear_model import Lasso
from sklearn.model_selection import
train_test_split
from sklearn.metrics import
mean_squared_error, r2_score
data = load_diabetes()
X = data.data
Y = data.target
X_train, X_test, Y_train, Y_test = train_test_split(
        X, Y, test_size = 0.2, random_state = 42)
lasso = Lasso(alpha = 0.1)
lasso.fit(X_train, Y_train)
Y_pred = lasso.predict(X_test)
print("simple Lasso Regression")
point("MSE:", mean_squared_error(Y_test, Y_pred))
print("R² Score:", r2_score(Y_test, Y_pred))
print("coefficients (first 5):", lasso.coef_[:5])
print("Intercept:", lasso.intercept_)
```

OUTPUT

```
Simple Lasso Regression
MSE: 2798.193485169719
R² Score: 0.4718547867276227
Coefficients (first 5): [0. -152.66477923  552.69777529
  803.36515791  -81.36500664]
Intercept: 151.57485282893947
```